

University of Agriculture, Fsd

Lab Submission

Software Configuration Management - Lab (SE 503)

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Instructor

Semester : Fall 2024



Lab # 1

Introduction to GitHub for Configuration Management

Task:

1. Sign Up a GitHub account

Already have an account? [Sign in →](#)

Sign up to GitHub

Email*

mahmadtayyab@gmail.com

Password*

.....

Password should be at least 15 characters OR at least 8 characters including a number and a lowercase letter.

Username*

Ahmad-tayyab-11|

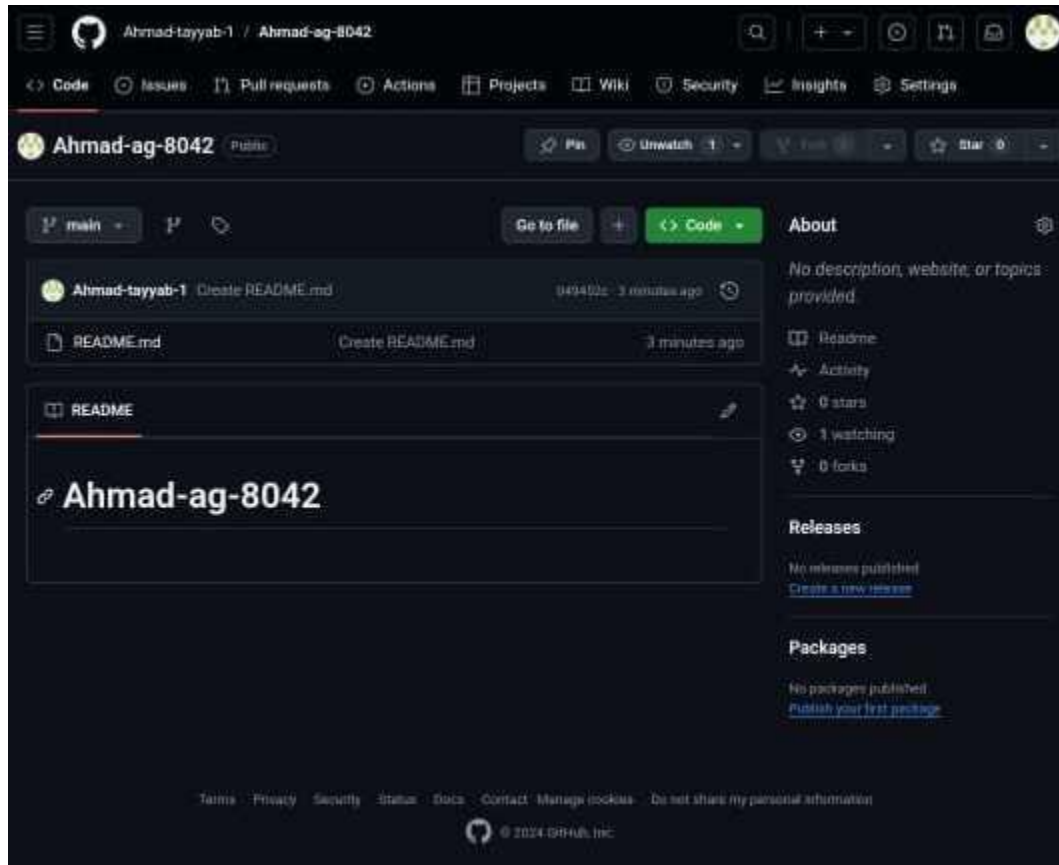
Ahmad-tayyab-11 is available.

Username may only contain alphanumeric characters or single hyphens, and cannot begin or end with a hyphen.

By creating an account, you agree to the [Terms of Service](#). For more information about GitHub's privacy practices, see the [GitHub Privacy Statement](#). We'll occasionally send you account-related emails.

Continue >

2. Create a repository named **Ahmad-**

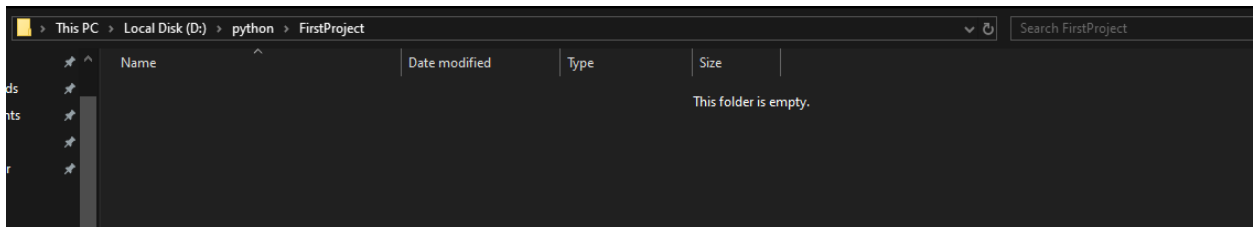


Lab # 2

Setting Up a Development Environment and Working with Git

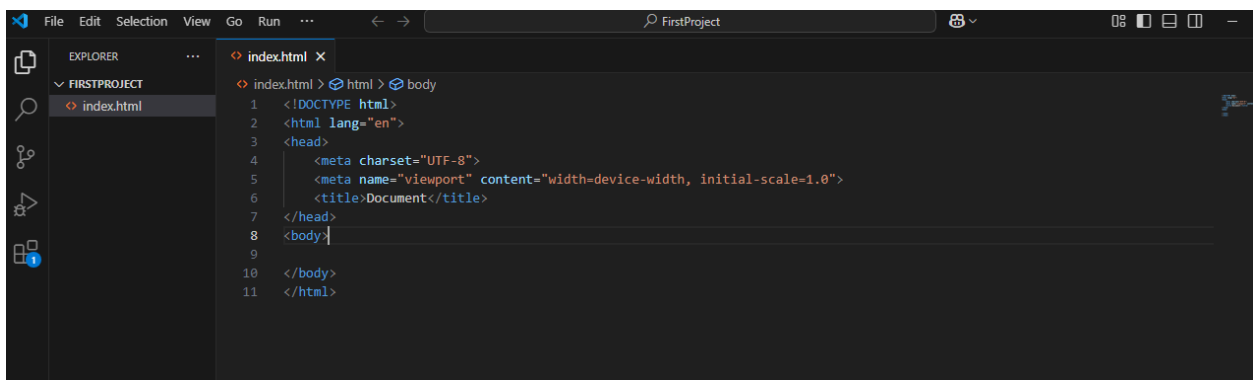
Task 1: Creating and Managing Commits.

- Create a New Local Repository:



- Make changes and stage files

- Commit the changes



```
2 <html lang="en">
3 <head>
4 <meta charset="UTF-8">

PROBLEMS OUTPUT TERMINAL PORTS DEBUG CONSOLE powershell
● PS D:\python\FirstProject> git init -b main
Initialized empty Git repository in D:/python/FirstProject/.git/
● PS D:\python\FirstProject> git add .
● PS D:\python\FirstProject> git commit -m "intial commit For First Project"
[main (root-commit) 27a3f2f] intial commit For First Project
1 file changed, 11 insertions(+)
create mode 100644 index.html
○ PS D:\python\FirstProject>
```

Using the **git log** cmd to check commit history to verify

```
● PS D:\python\FirstProject> git branch -M main
● PS D:\python\FirstProject> git push -u origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 4 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 379 bytes | 189.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/Ahmad-tayyab-1/FirstProject.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
○ PS D:\python\FirstProject>
```

```
● PS D:\python\FirstProject> git log
commit 27a3f2f7e4e02029c5d941054c732ba436815776 (HEAD -> main, origin/main)
Author: Ahmad-tayyab-1 <mahmadtayyab45@gmail.com>
Date: Thu Jan 2 13:25:24 2025 +0500

    intial commit For First Project
○ PS D:\python\FirstProject>
```

```
PS D:\python\FirstProject> git checkout -b Second-branch
Switched to a new branch 'Second-branch'
PS D:\python\FirstProject> git add index.html
```

Use **git add** and **git commit -m** to make changes in the new branch

3. Switch Back to the Master Branch

```
no changes added to commit (use "git add" and/or "git commit -a")
PS D:\python\FirstProject> git add index.html
PS D:\python\FirstProject> git commit -m "adding files to new branch"
[Second-branch c122572] adding files to new branch
1 file changed, 1 insertion(+), 1 deletion(-)
PS D:\python\FirstProject>
```

4. Merge the Feature Branch into Master

Use **git merge <branch-name>** to merge the new branch to main branch

```
PS D:\python\FirstProject> git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
PS D:\python\FirstProject>
```

5. Push the Merged Changes to GitHub

Use **git push -u origin main** to push the changes to github

```
PS D:\python\FirstProject> git merge Second-branch
Updating 27a3f2f..c122572
Fast-forward
 index.html | 2 +-
1 file changed, 1 insertion(+), 1 deletion(-)
PS D:\python\FirstProject>
```

```
Fast-forward
 index.html | 2 +-
1 file changed, 1 insertion(+), 1 deletion(-)
PS D:\python\FirstProject> git push -u origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 4 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 311 bytes | 311.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/Ahmad-tayyab-1/FirstProject.git
27a3f2f..c122572 main -> main
branch 'main' set up to track 'origin/main'.
PS D:\python\FirstProject>
```

Lab # 3

Advanced GitHub Features

Task 1: Creating and Managing Issues

1. Create an Issue:

-
-
-
-

Open a repository on GitHub.

Navigate to the Issues tab and click on New Issue.

Fill in the issue title and

description.

Ahmad-tayyab-1 / Ahmad-ag-8042

Type to search

Code Issues Pull requests Discussions Actions Projects Wiki Security Insights Settings

Add a title

Fix typo in README.md

Add a description

Write Preview

There is a typo in the forth line of the README file. Update it

Markdown is supported Paste, drop, or click to add files

Assignees

No one—assign yourself

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Shows branches and pull requests linked to this issue.

Helpful resources

GitHub Community Guidelines

Submit new issue

Remember, contributions to this repository should follow our GitHub Community Guidelines.

3. Close the

○

○

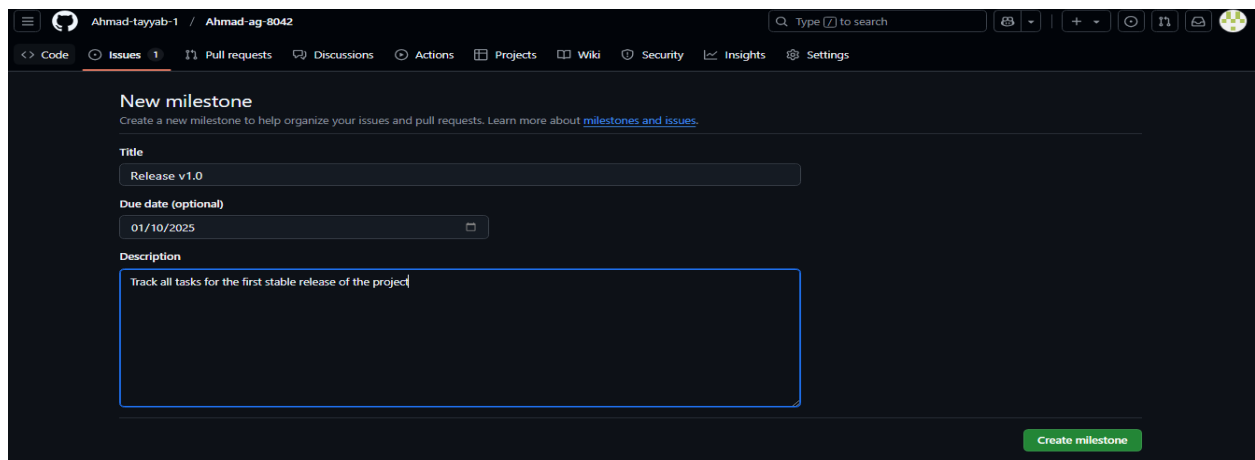
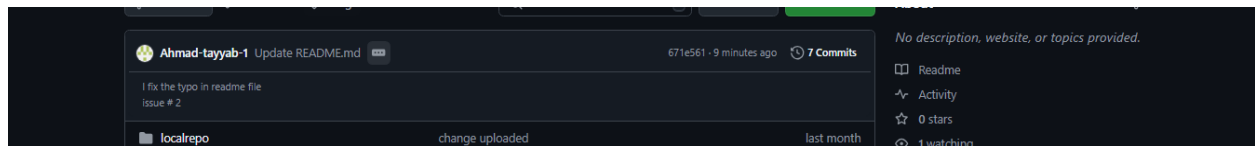
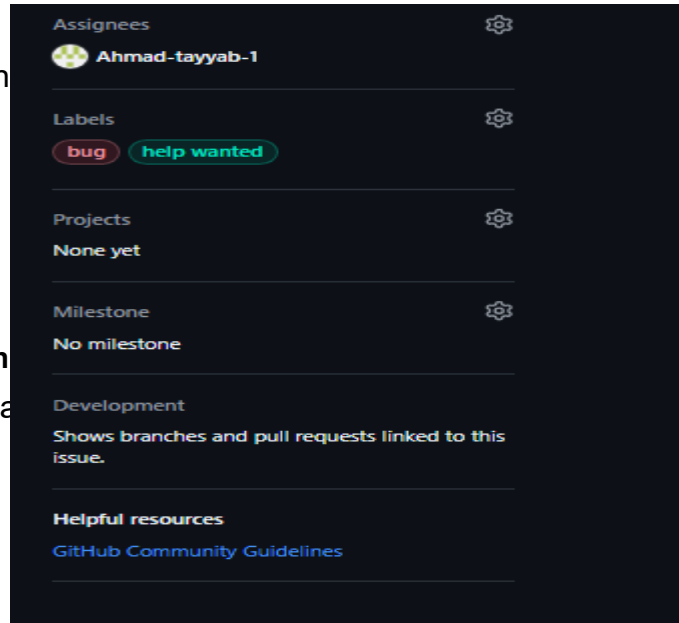
a commit by including

Task 2: Planning

1. Create a

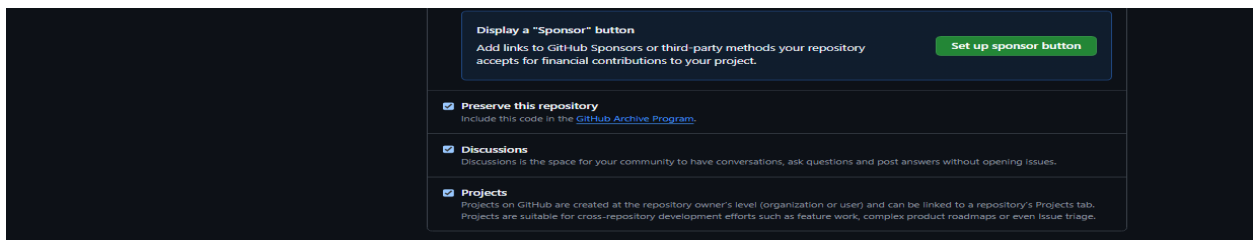
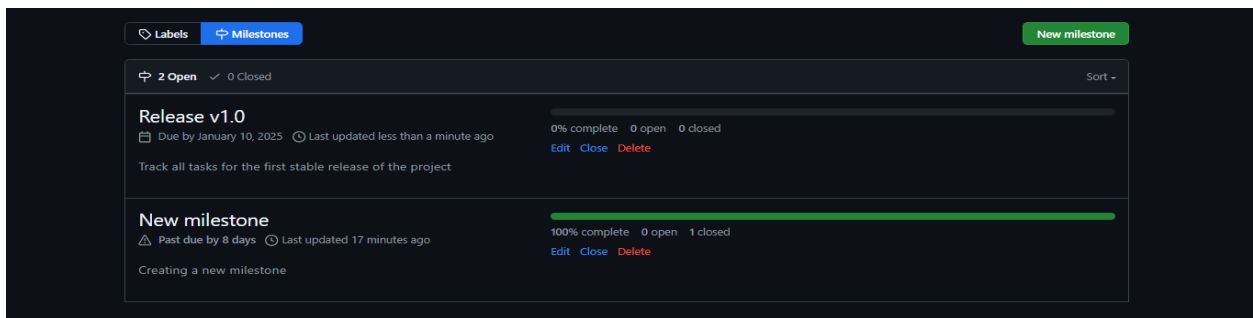
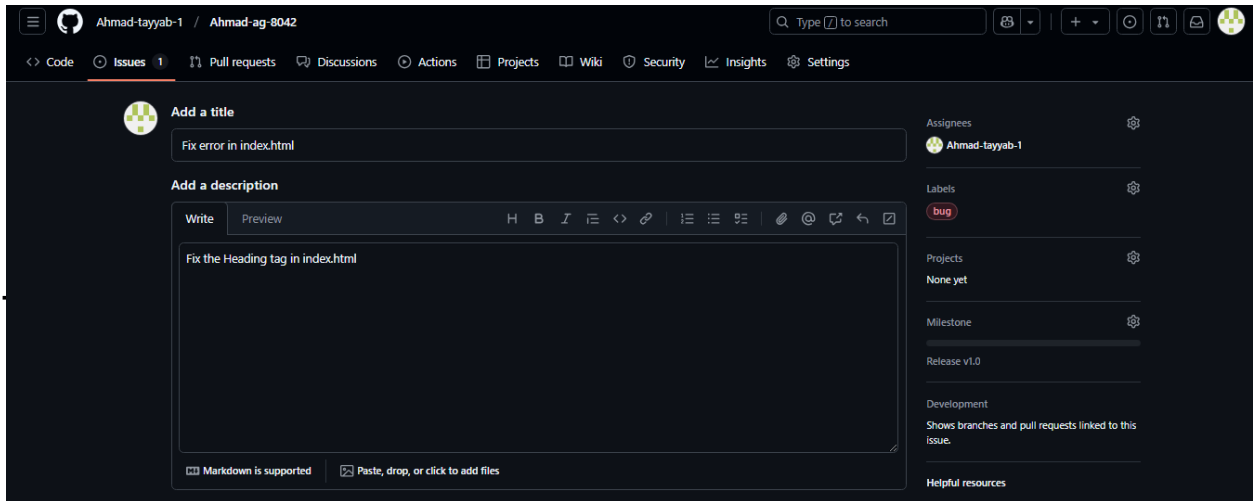
○

stone.



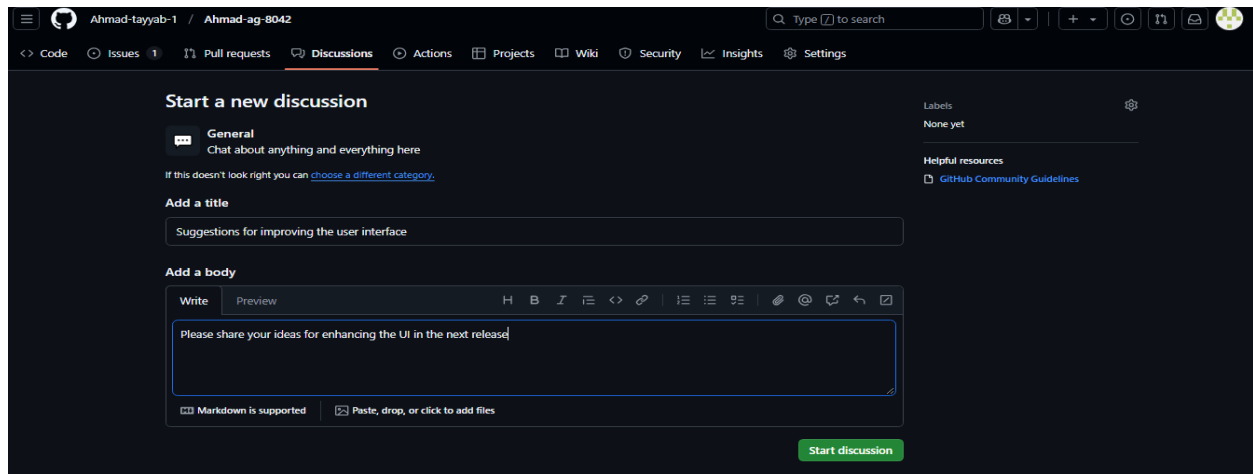
2.Add Issues to Milestone:

- Open an issue and assign it to the milestone.
- Use multiple issues to plan tasks related to the milestone.



2.Start a Discussion:

- Navigate to the Discussions tab and click New Discussion.
- Select the discussion category (e.g., General, Q&A).



The screenshot shows the GitHub interface for starting a new discussion. The top navigation bar includes links for Code, Issues, Pull requests, Discussions (selected), Actions, Projects, Wiki, Security, Insights, and Settings. The main heading is 'Start a new discussion'. Below it, the 'General' category is selected with the description 'Chat about anything and everything here'. A link to 'choose a different category' is provided. The 'Add a title' field contains the text 'Suggestions for improving the user interface'. The 'Add a body' section has a rich text editor with the text 'Please share your ideas for enhancing the UI in the next release'. A 'Start discussion' button is at the bottom right.



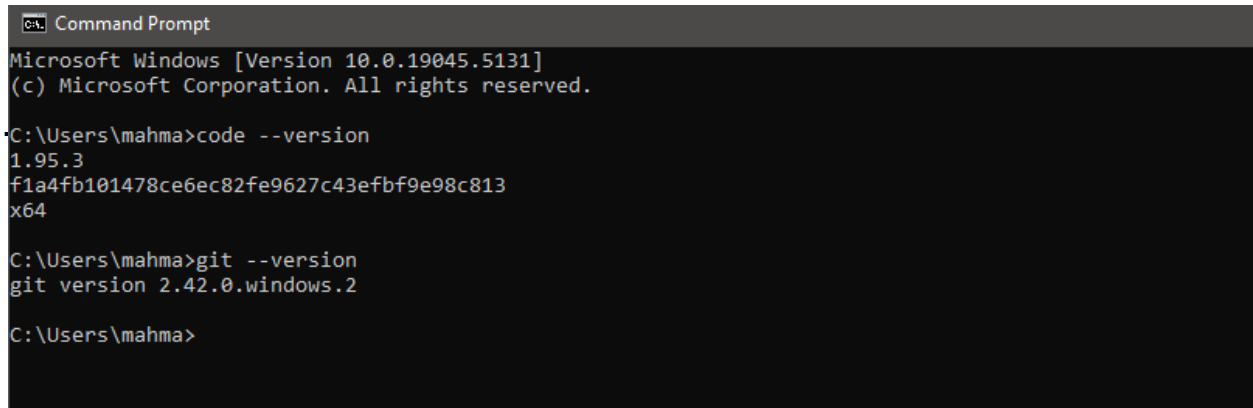
The screenshot shows a GitHub discussion thread titled 'Suggestions for improving the user interface #6'. The thread was started by 'Ahmad-tayyab-1' 1 minute ago. The main post contains the text 'Please share your ideas for enhancing the UI in the next release' and has 1 upvote. A comment by 'Ahmad-tayyab-1' is shown, titled '10 Tips How To Improve UI/UX Design of a Web Application'. The comment lists ten tips: Know Your Audience, Plan and Prioritize Content, Implement Responsive Design, Optimize Page Load Speed, Ensure Design Consistency, Prioritize Readability and Accessibility, Streamline User Inputs, and Visual Feedback and Interactivity. The right sidebar shows the category 'General', labels 'None yet', 1 participant, and a notification to 'Unsubscribe'. At the bottom, there are options to lock, transfer, pin, create issue from, or delete the discussion.

Lab # 4

Setting Up a Development Environment and Working with Git

Task 1: Download and Install VS Code and Git

- Downloaded Visual Studio Code from the official website.
- Downloaded Git from the official Git website.
- verified the installation using commands like **git --version** and **code --version**

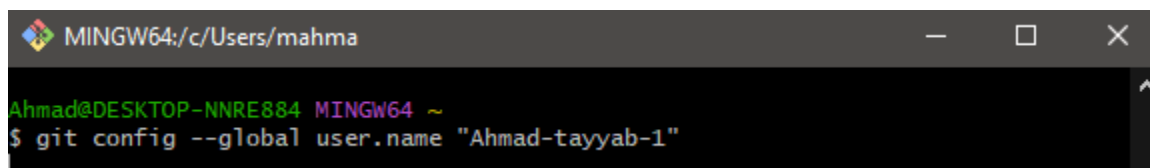


```
C:\> Command Prompt
Microsoft Windows [Version 10.0.19045.5131]
(c) Microsoft Corporation. All rights reserved.

C:\Users\mahma>code --version
1.95.3
f1a4fb101478ce6ec82fe9627c43efbf9e98c813
x64

C:\Users\mahma>git --version
git version 2.42.0.windows.2

C:\Users\mahma>
```

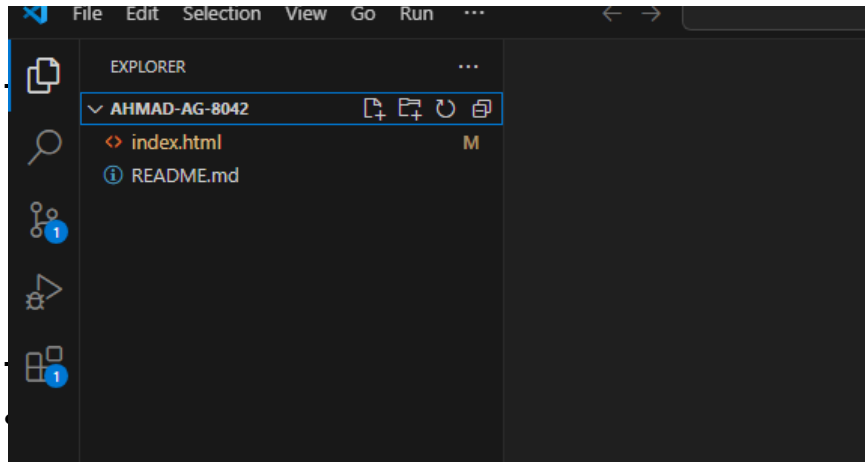


```
MINGW64:/c/Users/mahma
Ahmad@DESKTOP-NNRE884 MINGW64 ~
$ git config --global user.name "Ahmad-tayyab-1"
```

```
MINGW64:/c/Users/mahma
Ahmad@DESKTOP-NNRE884 MINGW64 ~
$ git config --global user.email "mahmadtayyab45@gmail.com"
Ahmad@DESKTOP-NNRE884 MINGW64 ~
```

Task 4: Create a Folder and Open in VS Code:

```
MINGW64:/c/Users/mahma
Ahmad@DESKTOP-NNRE884 MINGW64 ~
$ git config --global user.email "mahmadtayyab45@gmail.com"
Ahmad@DESKTOP-NNRE884 MINGW64 ~
$ git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=openssl
http.sslcainfo=C:/Program Files/Git/mingw64/etc/ssl/certs/ca-bundle.crt
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
credential.helper=manager
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=master
user.name=Ahmad-tayyab-1
user.email=mahmadtayyab45@gmail.com
Ahmad@DESKTOP-NNRE884 MINGW64 ~
$ |
```



ne <repo link>

changes, and saved them.

Task 7: Check Git Status:

- Use the **git status** command to check the status of files.
- Observed the file states such as untracked, modified, staged, or deleted.

```
PS D:\python\SE-503\Ahmad-ag-8042> git clone https://github.com/Ahmad-tayyab-1/Ahmad-ag-8042.git
Cloning into 'Ahmad-ag-8042'...
remote: Enumerating objects: 9, done.
remote: Counting objects: 100% (9/9), done.
remote: Compressing objects: 100% (5/5), done.
remote: Total 9 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (9/9), done.
PS D:\python\SE-503\Ahmad-ag-8042>
```

```
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   index.html

no changes added to commit (use "git add" and/or "git commit -a")
PS D:\python\SE-503\Ahmad-ag-8042>
```

Lab # 5

Version Control with Git: Managing Changes, Uploading Projects to GitHub, and Working with Branches

Task 1: Stage and Commit Changes

1. Open the terminal in the cloned repository folder (from the previous lab).

```
PS D:\python\SE-503\Ahmad-ag-8042>
* History restored
PS D:\python\SE-503\Ahmad-ag-8042>
```

3. Stage the modified or new files:

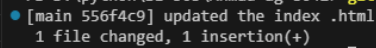
```
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   index.html

no changes added to commit (use "git add" and/or "git commit -a")
PS D:\python\SE-503\Ahmad-ag-8042>
```

Task 2: Push Changes to GitHub

```
PS D:\python\SE-503\Ahmad-ag-8042> git commit -m "updated the index .html "
```

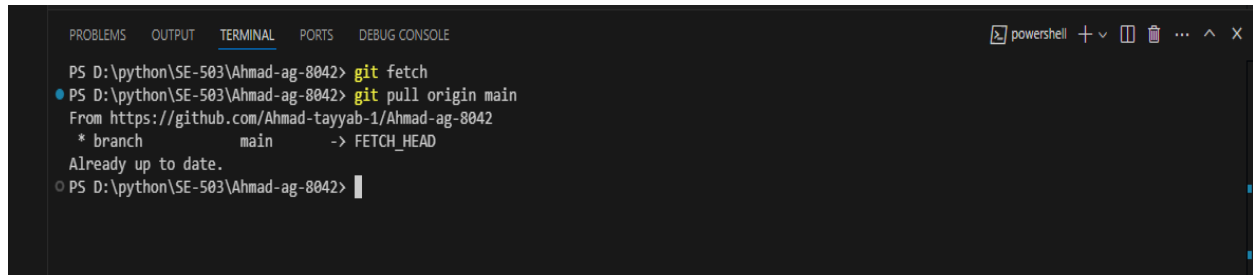


```
● [main 556f4c9] updated the index .html
  1 file changed, 1 insertion(+)
○ PS D:\python\SE-503\Ahmad-ag-8042>
```

1. If someone has updated the repository on Git prevents you from pushing to avoid overwriting others. Execute the following commands to update the local repo:

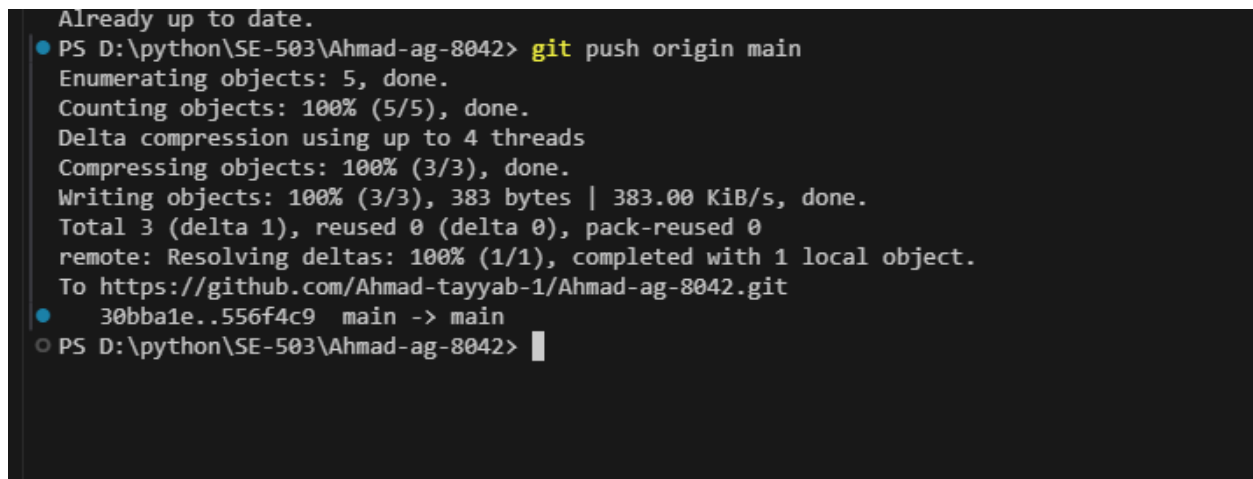
git fetch origin

git pull origin <branch_name>

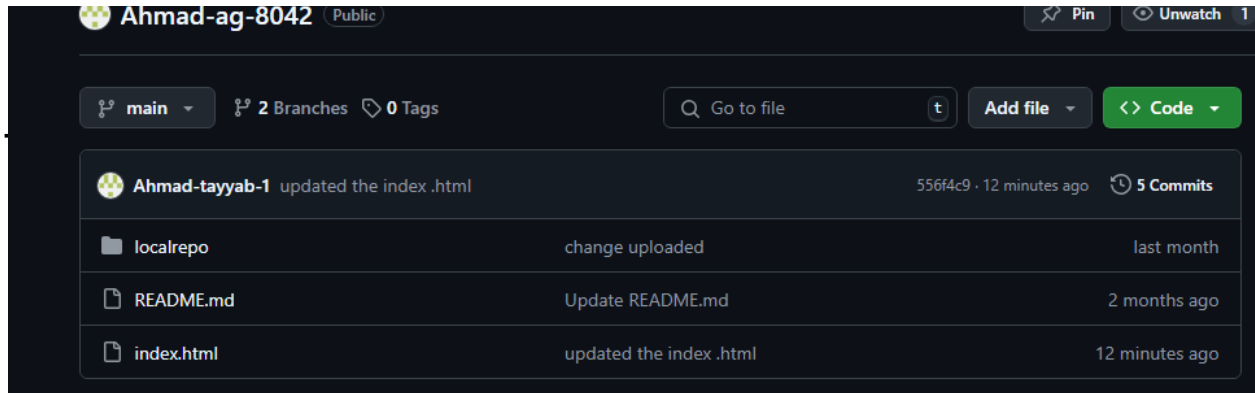
A screenshot of a PowerShell terminal window. The window has a title bar with 'powershell' and standard window controls. The terminal shows the following commands and output:

```
PS D:\python\SE-503\Ahmad-ag-8042> git fetch
PS D:\python\SE-503\Ahmad-ag-8042> git pull origin main
From https://github.com/Ahmad-tayyab-1/Ahmad-ag-8042
* branch      main      -> FETCH_HEAD
Already up to date.
PS D:\python\SE-503\Ahmad-ag-8042>
```

3. Verify the changes on the GitHub repository by visiting the repository page in your browser

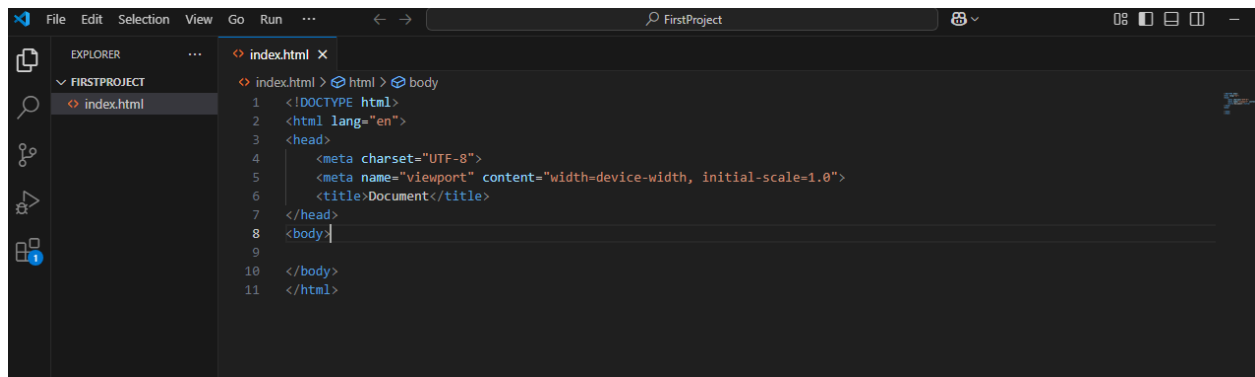
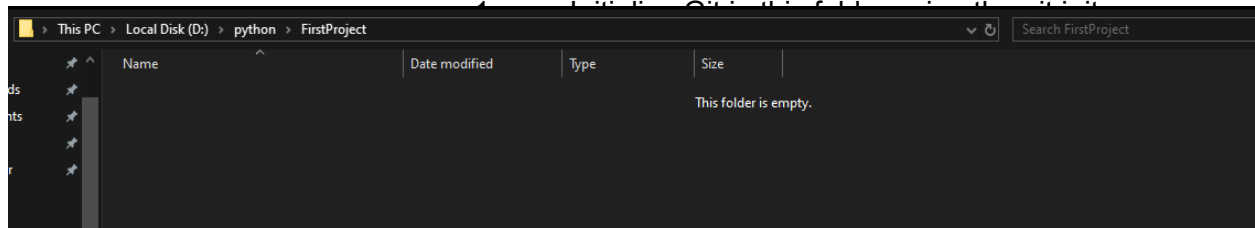
A screenshot of a PowerShell terminal window showing the output of a git push command. The terminal shows the following commands and output:

```
Already up to date.
PS D:\python\SE-503\Ahmad-ag-8042> git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 4 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 383 bytes | 383.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/Ahmad-tayyab-1/Ahmad-ag-8042.git
   30bba1e..556f4c9  main -> main
PS D:\python\SE-503\Ahmad-ag-8042>
```



2. Inside the folder, create files for your project (e.g., .txt, .html, .py, etc.).

Step 2: Initialize Git in the Folder




```
3 <head>
4 <meta charset="UTF-8">
```

PROBLEMS OUTPUT **TERMINAL** PORTS DEBUG CONSOLE

```
● PS D:\python\FirstProject> git init -b main
Initialized empty Git repository in D:/python/FirstProject/.git/
○ PS D:\python\FirstProject>
```

2. Commit the files with a descriptive message.

`git commit -m "Initial commit for FirstProject"`

Step 4: Create a New GitHub Repository

1. Log in to .
2. Click the **New** button to create a repository.
3. Provide a repository name (e.g.,

```
2 <html lang="en">
3 <head>
4 <meta charset="UTF-8">
```

PROBLEMS OUTPUT **TERMINAL** PORTS DEBUG CONSOLE powershell

```
● PS D:\python\FirstProject> git init -b main
Initialized empty Git repository in D:/python/FirstProject/.git/
● PS D:\python\FirstProject> git add .
● PS D:\python\FirstProject> git commit -m "initial commit For First Project"
[main (root-commit) 27a3f2f] initial commit For First Project
1 file changed, 11 insertions(+)
create mode 100644 index.html
○ PS D:\python\FirstProject>
```

Create a new repository

A repository contains all project files, including the revision history. Already have a project repository elsewhere? [Import a repository.](#)

Required fields are marked with an asterisk (*).

Owner * / Repository name *

Ahmad-tayyab-1 / FirstProject

FirstProject is available.

Great repository names are short and memorable. Need inspiration? How about [supreme-umbrella](#) ?

Description (optional)

☒ Public
Anyone on the internet can see this repository. You choose who can commit.

☐ Private
You choose who can see and commit to this repository.

Initialize this repository with:

☐ Add a README file
This is where you can write a long description for your project. [Learn more about READMEs.](#)

4. Check the current branch using **git branch**

Rename the branch from master to main (the convention github follows)

Using branch -M main

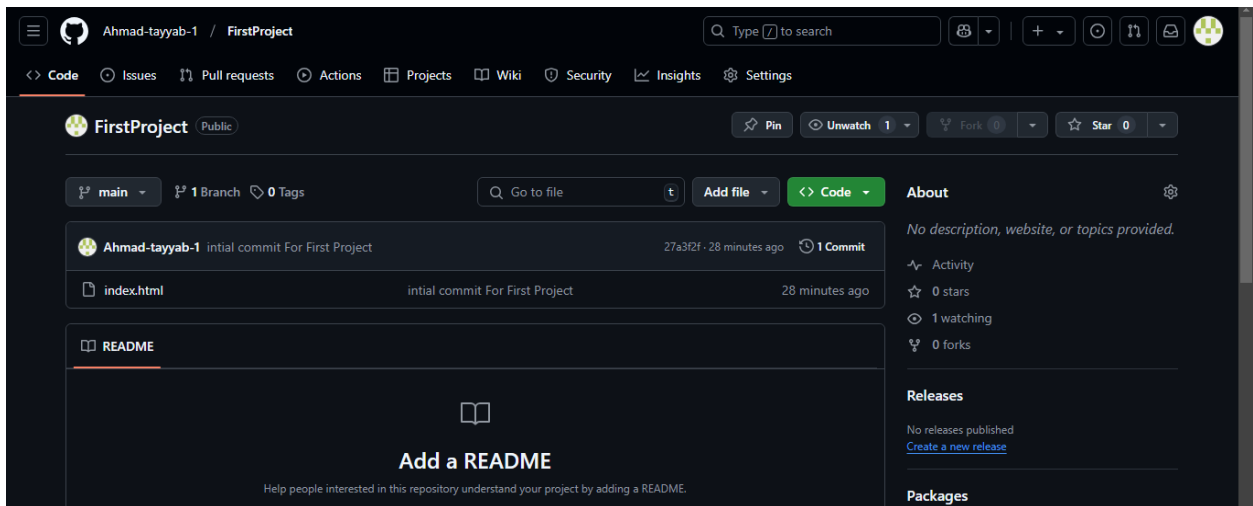
```
Create mode 100644 index.html
PS D:\python\FirstProject> git remote add origin https://github.com/Ahmad-tayyab-1/FirstProject.git
PS D:\python\FirstProject> 
```

```
PS D:\python\FirstProject> git remote add origin https://github.com/Ahmad-tayyab-1/FirstProject.git
PS D:\python\FirstProject> git remote -v
origin https://github.com/Ahmad-tayyab-1/FirstProject.git (fetch)
origin https://github.com/Ahmad-tayyab-1/FirstProject.git (push)
PS D:\python\FirstProject> 
```

```
origin = https://github.com/Ahmad-tayyab-1/FirstProject.git (push)
● PS D:\python\FirstProject> git branch
* main
● PS D:\python\FirstProject> git branch -M main
○ PS D:\python\FirstProject>
```

Verify the Upload Files on Github by opening the “FirstProject Repository” in your github account

```
● PS D:\python\FirstProject> git branch -M main
● PS D:\python\FirstProject> git push -u origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 4 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 379 bytes | 189.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/Ahmad-tayyab-1/FirstProject.git
 * [new branch]    main -> main
branch 'main' set up to track 'origin/main'.
○ PS D:\python\FirstProject>
```



Lab # 6

Working with Branches: Creating, Modifying, Merging, Simulating, and Resolving Conflicts

Task 1 :Resolving Merge Conflict

- Create a new branch “**feature1**”

Using **git branch feature1**

- Navigate to **feature1** branch

Using **git checkout feature1**

```
fatal: a branch named 'feature1' already exists
PS D:\python\FirstProject> git branch feature-1
PS D:\python\FirstProject> git checkout feature1
Switched to branch 'feature1'
PS D:\python\FirstProject>
```

Using **git add index.html** and **git commit -m 'feature1 edit'**

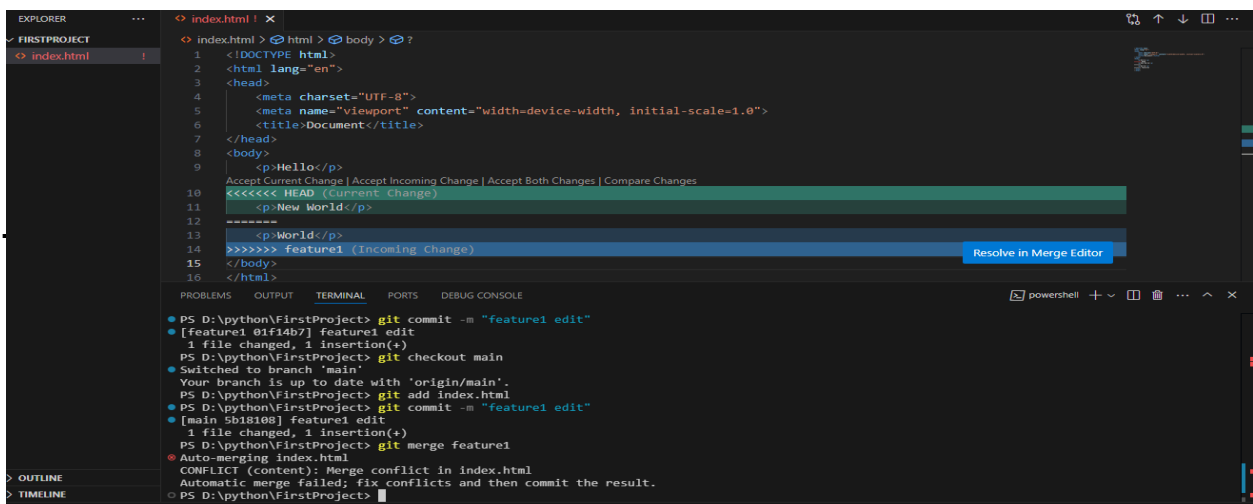
- Navigate back to main branch and make changes in the same file as feature1 branch

```
PS D:\python\FirstProject> git checkout feature1
Switched to branch 'feature1'
PS D:\python\FirstProject> git add index.html
1 file changed, 1 insertion(+)
```

- Merge the feature1 branch to main branch

```
1 file changed, 1 insertion(+)
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
PS D:\python\FirstProject> git add index.html
PS D:\python\FirstProject> git commit -m "feature1 edit"
[main 5b18108] feature1 edit
1 file changed, 1 insertion(+)
PS D:\python\FirstProject> git merge feature1
Auto-merging index.html
CONFLICT (content): Merge conflict in index.html
```

- Resolve the merge conflict by manually



```
PS D:\python\FirstProject> git add index.html
PS D:\python\FirstProject> git commit -m "accept the feature branch change"
[main a05cd27] accept the feature branch change
PS D:\python\FirstProject> git merge feature1
```

```
index.html M
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4 | /meta charset="UTF-8" \

PROBLEMS OUTPUT TERMINAL PORTS DEBUG CONSOLE
PS D:\python\FirstProject> git add index.html
PS D:\python\FirstProject>
```

- Using **git commit -m "New h1 tag added"**

3. Undo the last commit while keeping the changes:

```
3 <head>
4 | /meta charset="UTF-8" \

PROBLEMS OUTPUT TERMINAL PORTS DEBUG CONSOLE
PS D:\python\FirstProject> git add index.html
PS D:\python\FirstProject> git commit -m "New h1 tag added"
[main 00bb703] New h1 tag added
1 file changed, 1 insertion(+)
PS D:\python\FirstProject>
```

```
PS D:\python\FirstProject> git commit -m "New h1 tag added"
[main 00bb703] New h1 tag added
1 file changed, 1 insertion(+)
PS D:\python\FirstProject> git reset --hard HEAD~1
HEAD is now at a05cd27 accept the feature branch change
PS D:\python\FirstProject>
```

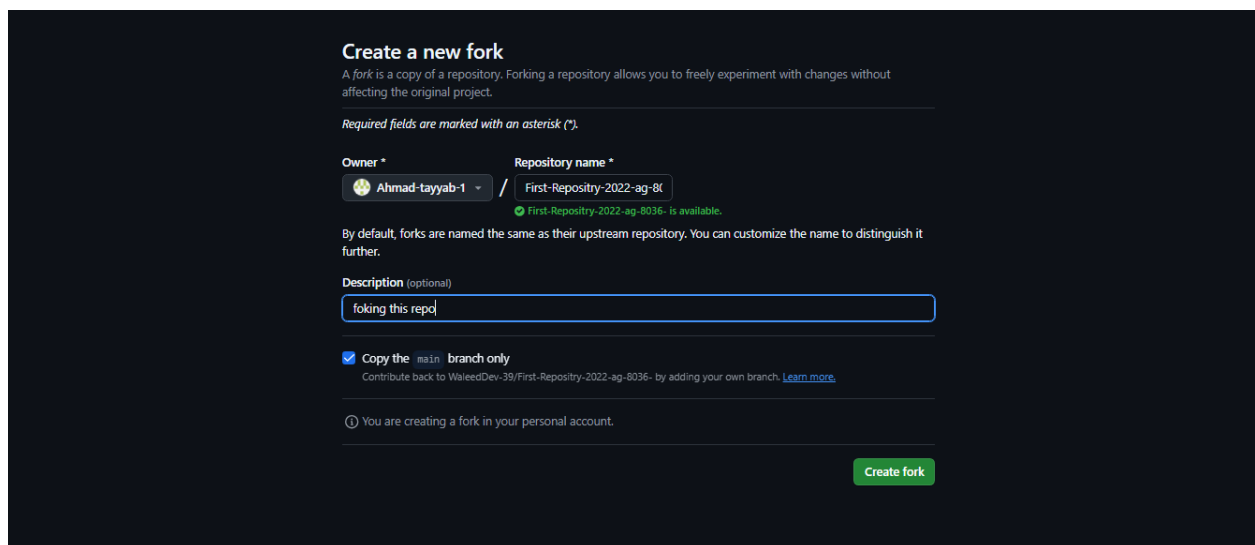
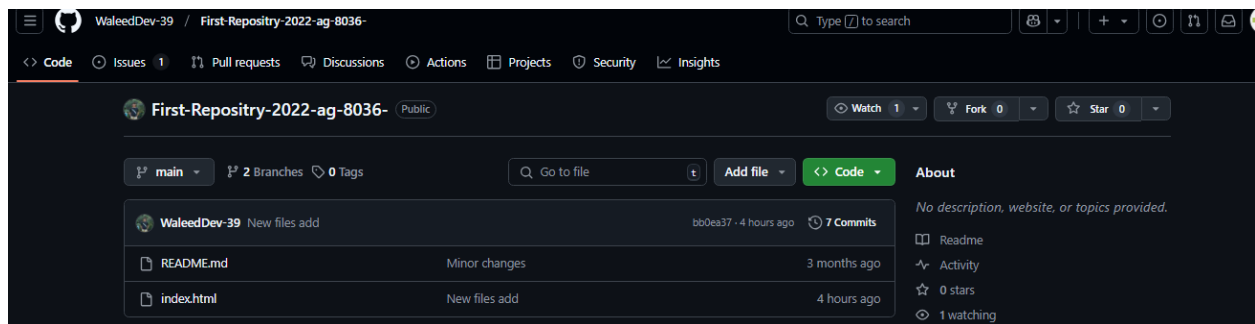
Lab # 7

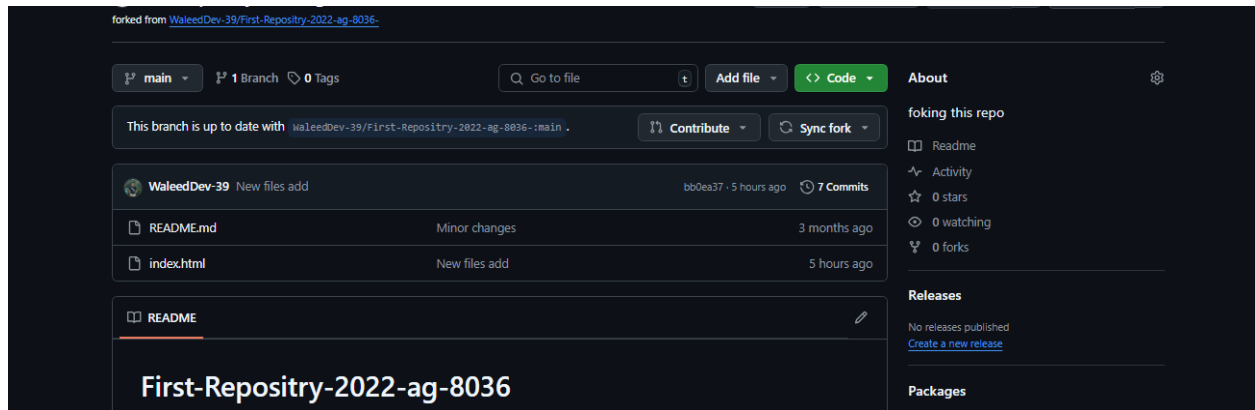
Advanced GitHub Features (Including Forking)

Task 1: Forking a Repository

1. Fork a Repository:

- Navigate to any public repository on GitHub (e.g., an open-source project or a provided practice repository).





3. Make Changes in Your Fork:

- Edit files or add new features in the forked repository

4. Commit and Push Changes:

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.19045.5247]
(c) Microsoft Corporation. All rights reserved.

D:\python\fork repo>git clone https://github.com/Ahmad-tayyab-1/First-Repository-2022-ag-8036-.git
Cloning into 'First-Repository-2022-ag-8036-'...
remote: Enumerating objects: 21, done.
remote: Counting objects: 100% (21/21), done.
remote: Compressing objects: 100% (15/15), done.
remote: Total 21 (delta 4), reused 10 (delta 3), pack-reused 0 (from 0)
Receiving objects: 100% (21/21), 4.88 KiB | 499.00 KiB/s, done.
Resolving deltas: 100% (4/4), done.

D:\python\fork repo>
```


git add index.html

git commit -m "Added new tag in index.html"

```
PS D:\python\fork repo\First-Repository-2022-ag-8036-> git add index.html
PS D:\python\fork repo\First-Repository-2022-ag-8036-> git commit -m "added new html tag in index.html"
[main caa2861] added new html tag in index.html
1 file changed, 1 insertion(+), 4 deletions(-)
```

git push -u origin main

```
PS D:\python\fork repo\First-Repository-2022-ag-8036-> git add index.html
PS D:\python\fork repo\First-Repository-2022-ag-8036-> git commit -m "added new html tag in index.html"
[main caa2861] added new html tag in index.html
1 file changed, 1 insertion(+), 4 deletions(-)
PS D:\python\fork repo\First-Repository-2022-ag-8036-> git push -u origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 4 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 363 bytes | 363.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/Ahmad-tayyab-1/First-Repository-2022-ag-8036-.git
bb0ea37..caa2861 main -> main
branch 'main' set up to track 'origin/main'.
PS D:\python\fork repo\First-Repository-2022-ag-8036->
```

WaleedDev-39 / First-Repository-2022-ag-8036

Code Issues Pull requests Discussions Actions Projects Security Insights

Comparing changes

Choose two branches to see what's changed or to start a new pull request. If you need to, you can also [compare across forks](#) or [learn more about diff comparisons](#).

base repository: WaleedDev-39/First-Repository-... base: main head repository: Ahmad-tayyab-1/First-Repository-... compare: main

✓ Able to merge. These branches can be automatically merged.

added new html tag in index.html #6
Added a new paragraph tag in the index.html file [View pull request](#)

1 commit 1 file changed 1 contributor

Commits on Jan 2, 2025

added new html tag in index.html
Ahmad-tayyab-1 committed 12 minutes ago

Compare your changes with the original repository and submit the pull request.

- Add a **title** and **description** to explain your changes.

